Compressor off-design report: summary and diagnostics

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Rotor-Rotor Different Shaft  |  run: Max_eta_Compressor_design_rr_diff_shaft

Rotor-Rotor Different Shaft  |  PR_des = 1.5094, eta_des = 0.9589  |  correlations: polynomial, Howell factor 2.00

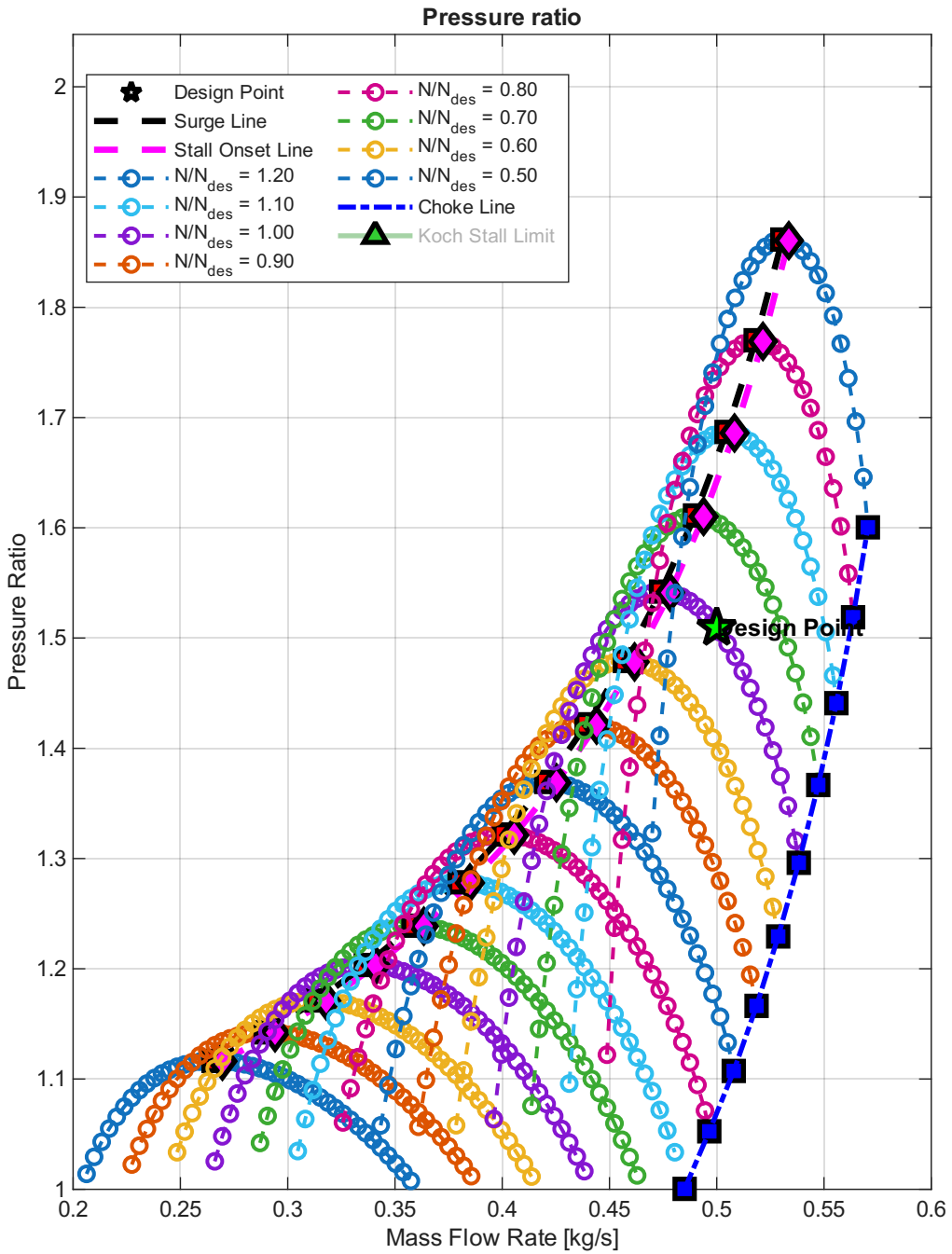
STABILITY (design speed): first limit = stall onset at m = 0.4783 kg/s (PR 1.541); stall margin SM = 6.7 %
   peak of the speed line (zero slope): m = 0.4741 kg/s | Howell stall onset: m = 0.4783 kg/s
   Koch stage limit: m = 0.5142 kg/s (indicative, see the Koch check below)
CHOKE (design speed): m_choke = 0.5383 kg/s (Rotor 1 exit (relative)) -> choke margin 7.7 %, operating range (m_choke - m_lim)/m_choke = 11.1 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.404 -> surge if the installation has V_p/(A_c L_c) > 3.92 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (stall onset): highest loss rise Rotor 1 (omega/omega_des = 2.00), lowest de Haller Rotor 1 (0.77). Details: Tables >
      Blade rows.

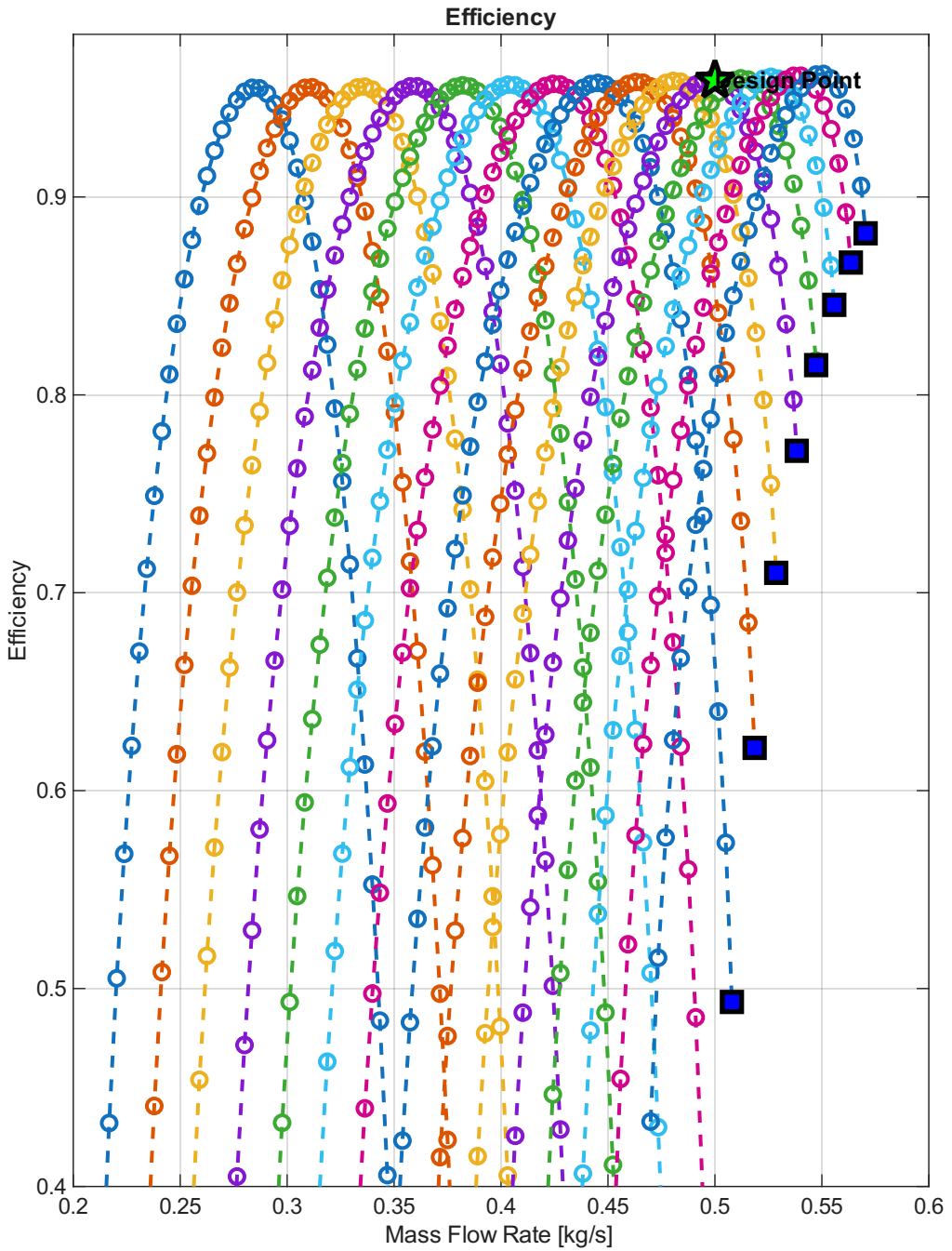
KOCH STAGE-LOADING INDICATOR (never a stability limit): NOT RECOMMENDED
   Not recommended for this design: it is outside the basis of Koch's correlation (see why). The Koch line is hidden by default; use Howell, the peak
      and the Blade rows table instead.
   Koch line: drawn on 15 of 15 speed lines (N/N_des 0.50 to 1.20).
   why: Rotors R1+R2: L/g2 = 0.54 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: R2: effective dynamic-head factor F_ef = 0.25, outside 0.39-1.11. The upstream wakes enter in a frame turning the same way at a similar
      speed; Koch's correlation does not cover this.
   why: Rotor-rotor stage: Koch's database holds rotor-stator stages only.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Rotors R1+R2: BEYOND THE LIMIT AT DESIGN (1.26)
     - RAISE the solidity x2.09 (stage L/g2 0.54 -> 1.12): R1 0.51 -> 1.06, R2 0.50 -> 1.05; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 33% at the same velocities (lower loading psi or degree of reaction, or a larger flow
       coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 1.16
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 1.18
     - R2: effective dynamic-head factor F_ef = 0.25 (upstream wakes enter in the same or a nearly equal frame and are not re-energized); with F_ef =
       1 the ratio would be 0.91 -> a stator between the rotors or a larger frame-speed difference (e.g. counter-rotation) helps
     - note: outside Koch's data: L/g2 = 0.54 (data 0.6-2.2), F_ef = [1.04 0.25] (data 0.39-1.11) -> indicative only
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Inputs of this run (.mat data and settings)

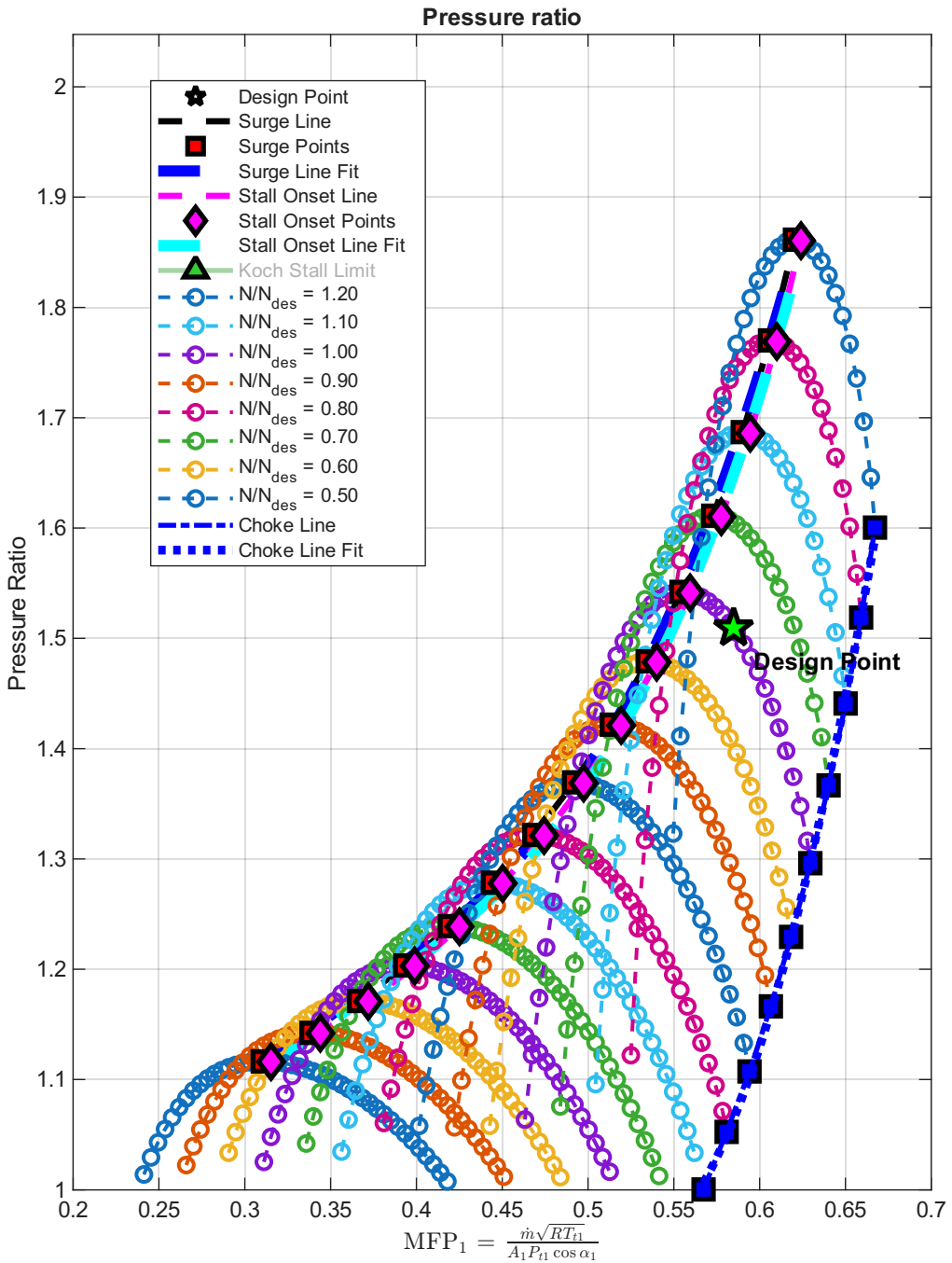
Variable	Value	Unit	Source
Configuration	Rotor-Rotor different shaft		
Input source	.mat: Max_eta_Compressor_design_rr_...		
alpha1	34.0366	deg	.mat: Max_eta_Compressor_design_rr_...
n_stages	2	-	.mat: Max_eta_Compressor_design_rr_...
A_vec	0.00296617 0.00273918 0.00265316	m^2	.mat: Max_eta_Compressor_design_rr_...
beta1_dash	53.5959	deg	.mat: Max_eta_Compressor_design_rr_...
beta2_dash	21.4107	deg	.mat: Max_eta_Compressor_design_rr_...
m_dot	0.5	kg/s	.mat: Max_eta_Compressor_design_rr_...
w_loss_rotor	0.0342878	-	.mat: Max_eta_Compressor_design_rr_...
beta3_dash	35.0314	deg	.mat: Max_eta_Compressor_design_rr_...
beta4_dash	0.389282	deg	.mat: Max_eta_Compressor_design_rr_...
deviation_1	12.6259	deg	.mat: Max_eta_Compressor_design_rr_...
deviation_2	11.1621	deg	.mat: Max_eta_Compressor_design_rr_...
incidence_1	-6.42119	deg	.mat: Max_eta_Compressor_design_rr_...
incidence_2	-4.58787	deg	.mat: Max_eta_Compressor_design_rr_...
deflection_1	13.1381	deg	.mat: Max_eta_Compressor_design_rr_...
deflection_2	18.8922	deg	.mat: Max_eta_Compressor_design_rr_...
w_loss_rotor2	0.0286249	-	.mat: Max_eta_Compressor_design_rr_...
U1	293.36	m/s	.mat: Max_eta_Compressor_design_rr_...
U2	278.692	m/s	.mat: Max_eta_Compressor_design_rr_...
sigma_1	0.507386	-	.mat: Max_eta_Compressor_design_rr_...
sigma_2	0.500805	-	.mat: Max_eta_Compressor_design_rr_...
chord	0.01	m	.mat: Max_eta_Compressor_design_rr_...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	129	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

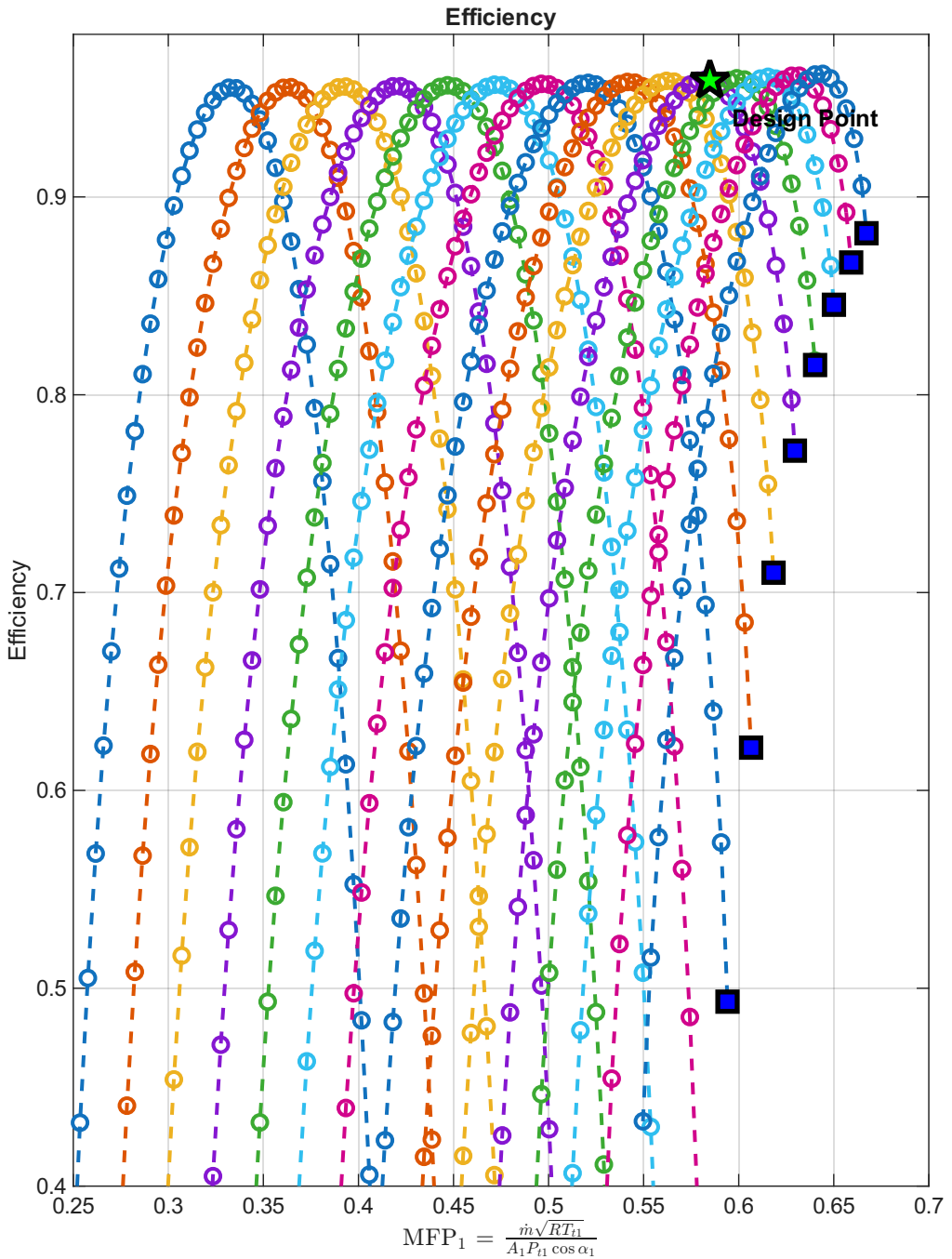
Standard (dimensional) map - Rotor-Rotor Different Shaft



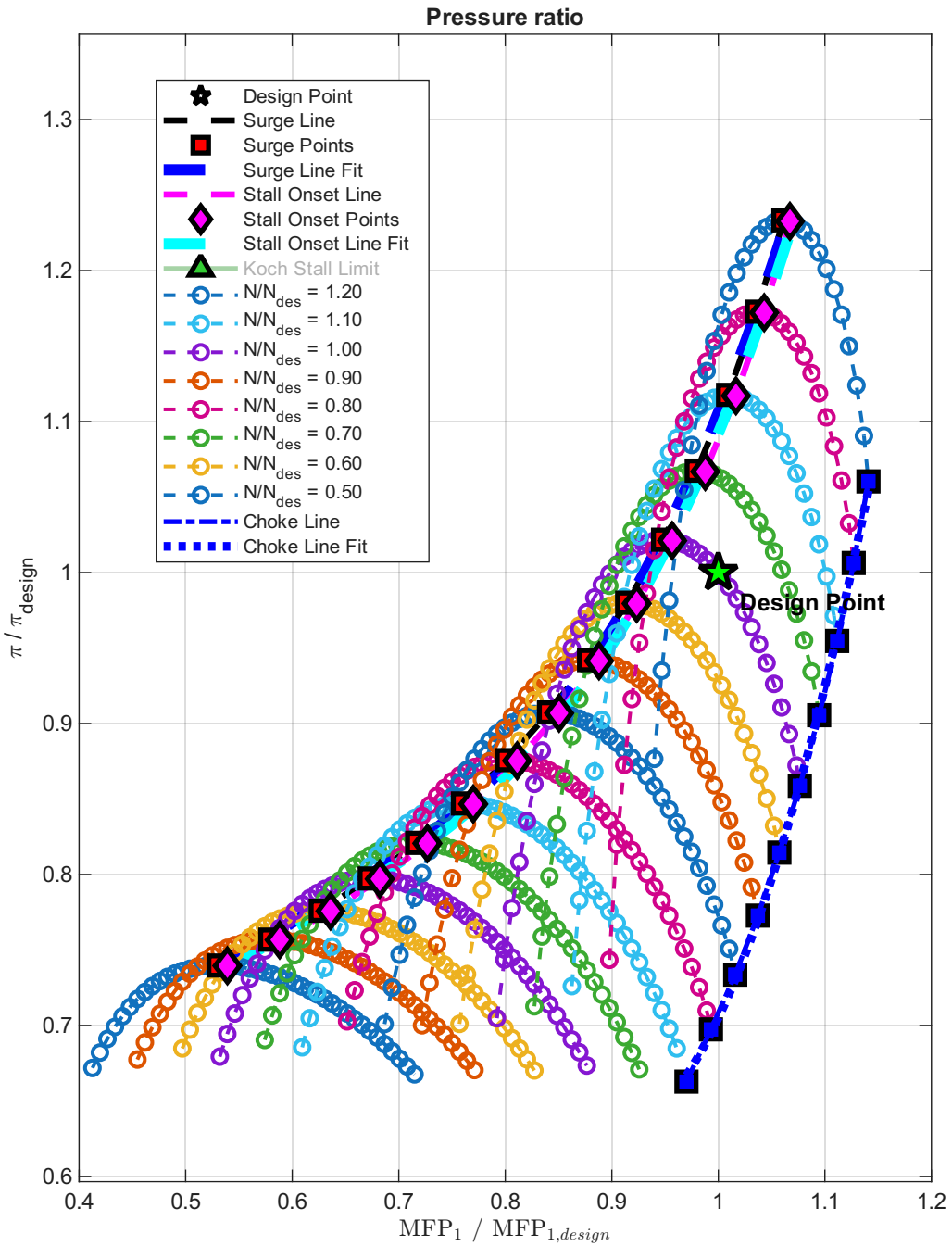


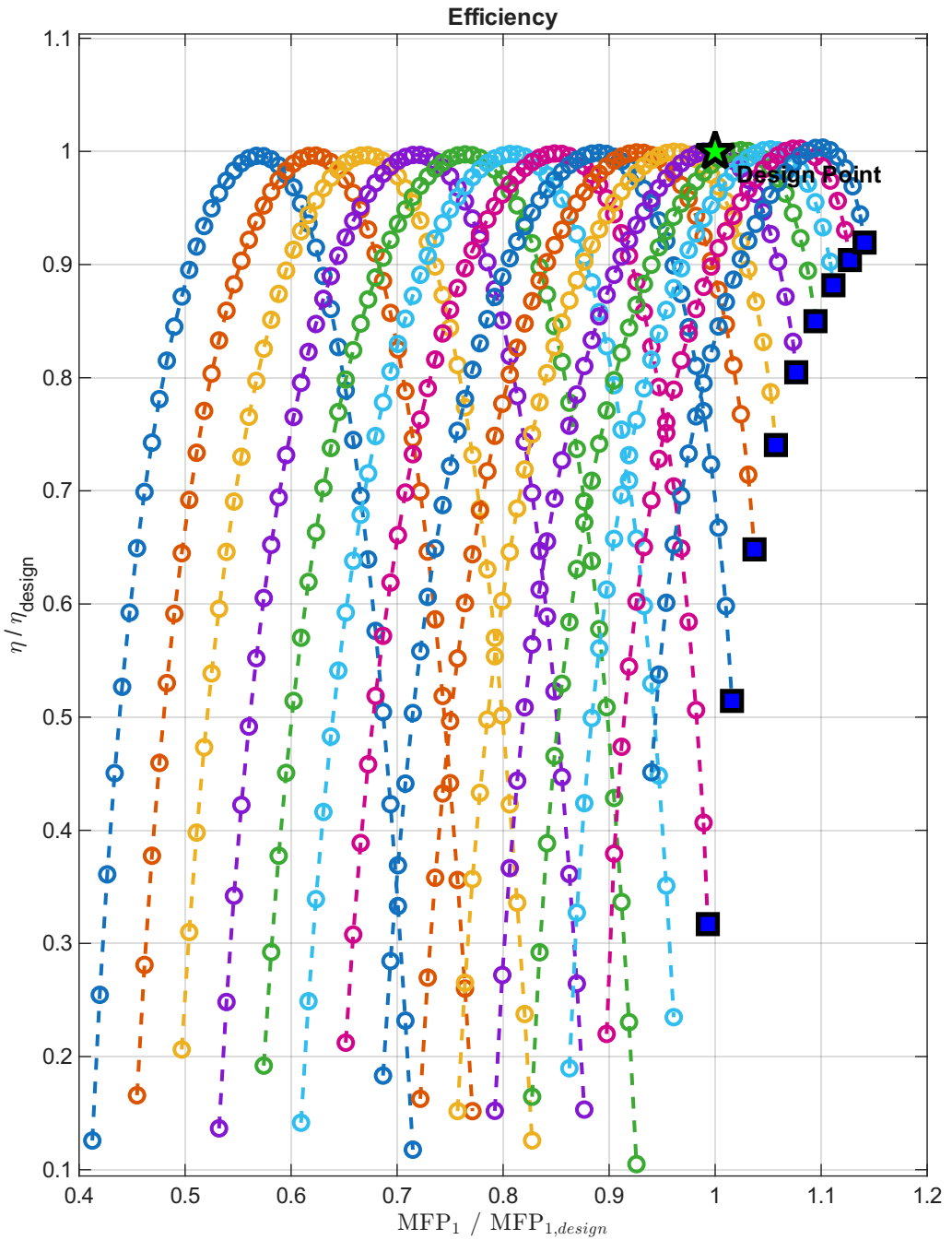
Non-dimensional map - Rotor-Rotor Different Shaft



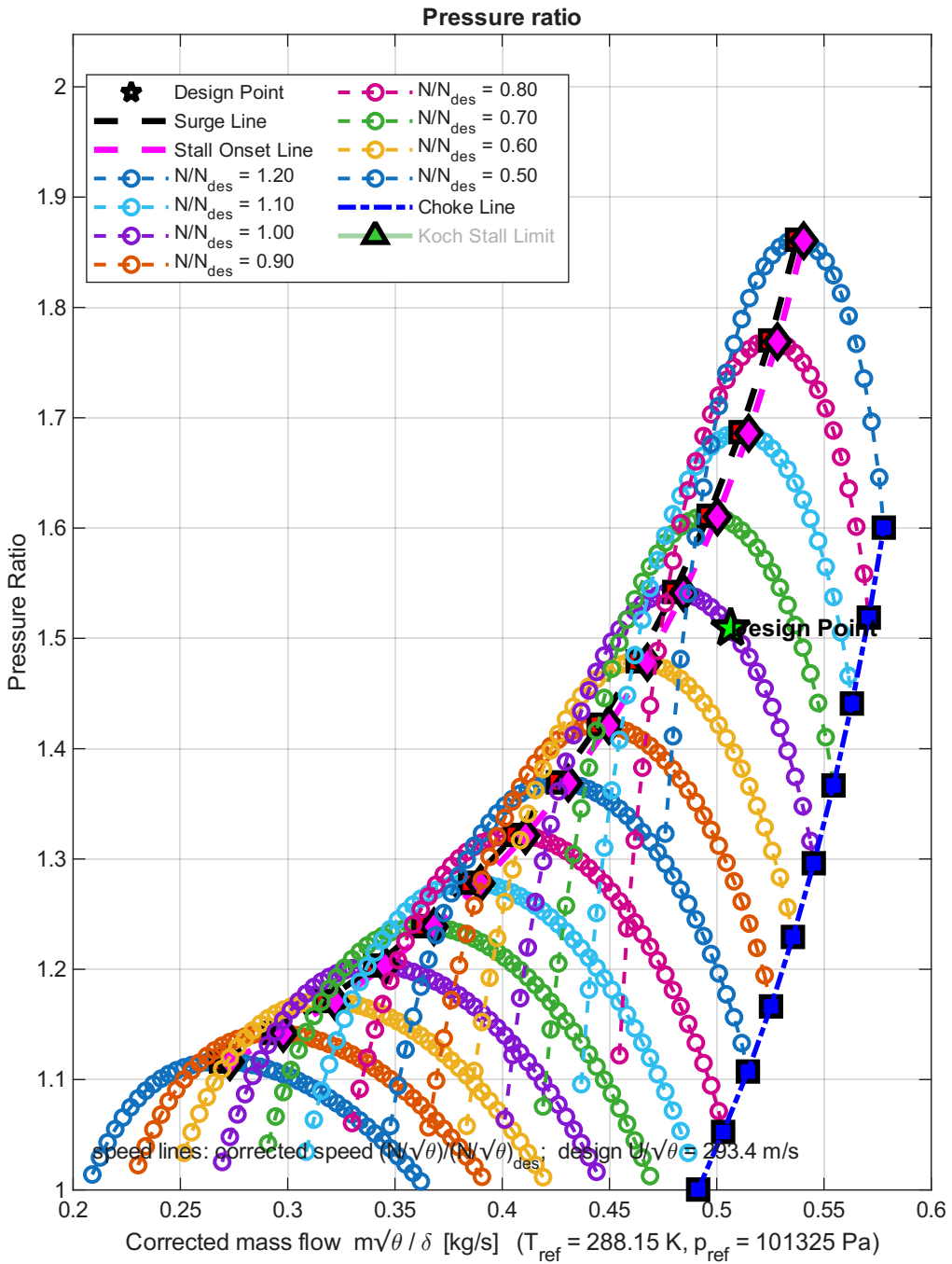


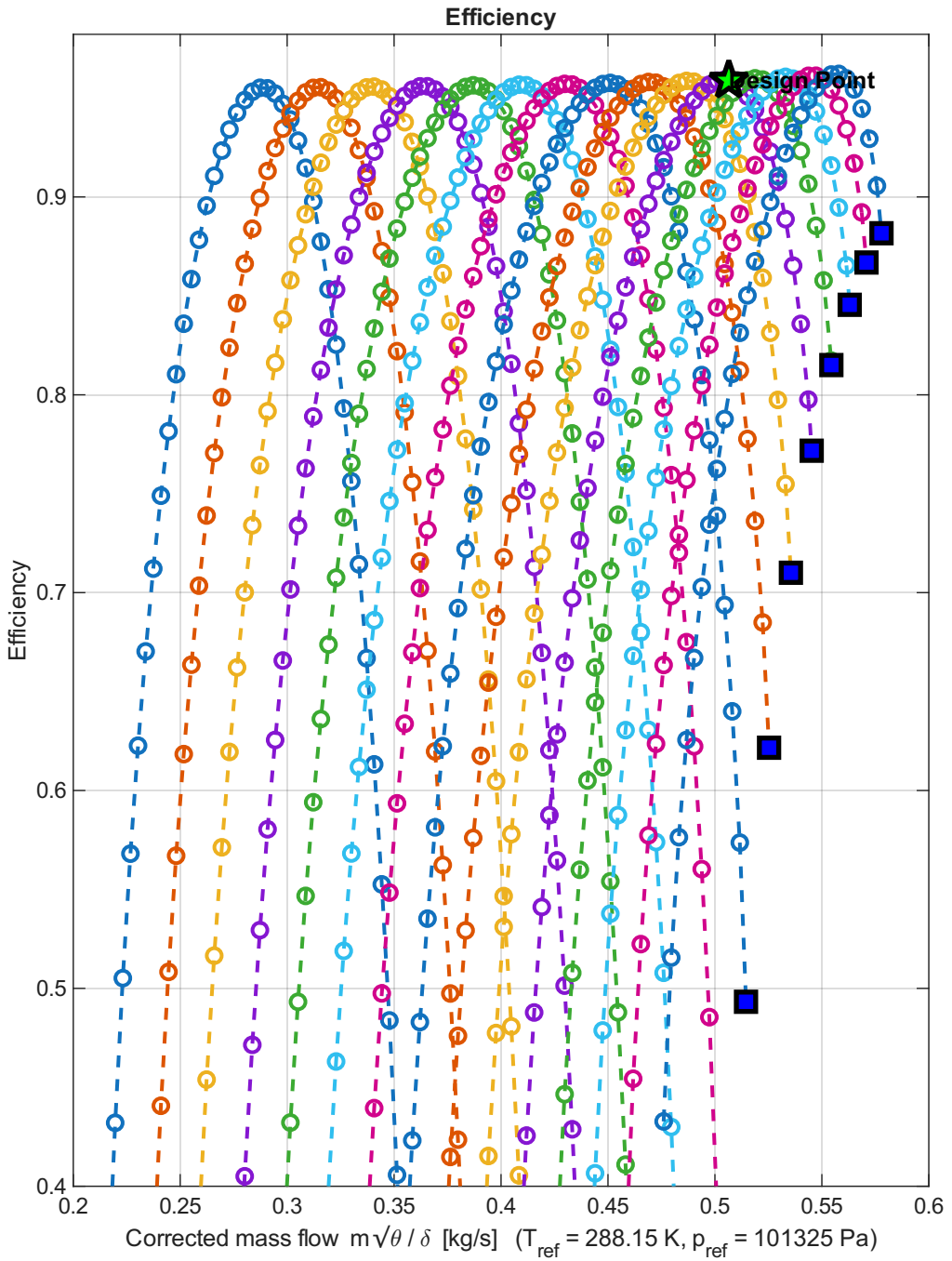
Relative non-dimensional map - Rotor-Rotor Different Shaft





Corrected map - Rotor-Rotor Different Shaft





Fitted boundary equations (surge, stall onset, choke)

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==== Non-dimensional Compressor Map: Pressure Ratio - Rotor-Rotor Different Shaft ====
Surge line:
Surge_PR = 23.3944*MFP^3 + -25.5813*MFP^2 + 10.4338*MFP + -0.3628   [R^2 = 0.99953]
Stall-onset line (Howell):
StallOnset_PR = 24.7951*MFP^3 + -27.6856*MFP^2 + 11.4005*MFP + -0.5092   [R^2 = 0.99945]
Choke line:
Choke_PR = 30.9943*MFP^2 + -32.4065*MFP + 9.4156   [R^2 = 0.99934]
==== Relative Non-dimensional Compressor Map: Pressure Ratio - Rotor-Rotor Different Shaft ====
Surge line:
(PR/PR_design) = 3.1002*x^3 + -5.7966*x^2 + 4.0427*x + -0.2404   (x = MFP/MFP_design)   [R^2 = 0.99953]
Stall-onset line (Howell):
(PR/PR_design) = 3.2858*x^3 + -6.2735*x^2 + 4.4172*x + -0.3374   (x = MFP/MFP_design)   [R^2 = 0.99945]
Choke line:
(PR/PR_design) = 7.0232*x^2 + -12.5563*x + 6.2381   (x = MFP/MFP_design)   [R^2 = 0.99934]
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Design point

Quantity	Value
Configuration	Rotor-Rotor Different Shaft
Pressure ratio PR_des	1.5094
Isentropic efficiency eta_des	0.95892
Mass flow m_des [kg/s]	0.5
MFP_1,des	0.58482
Corrected mass flow [kg/s]	0.50649
Inlet Tt1 [K] / Pt1 [Pa]	288 / 100000
Correlation set	polynomial
Howell factor	2

Choke

N/N_des	m_choke [kg/s]	MFP_1	PR	Eff	Choking row	Inside compressor region (PR>1)
0.5	0.42434	0.4963	0.788		Rotor 1 exit (relative)	no
0.55	0.43641	0.5104	0.824		Rotor 1 exit (relative)	no
0.6	0.44876	0.5249	0.863		Rotor 1 exit (relative)	no
0.65	0.46108	0.5393	0.906		Rotor 1 exit (relative)	no
0.7	0.47321	0.5535	0.951		Rotor 1 exit (relative)	no
0.75	0.48505	0.5673	1	0.003	Rotor 1 exit (relative)	yes
0.8	0.49654	0.5808	1.053	0.304	Rotor 1 exit (relative)	yes
0.85	0.50794	0.5941	1.107	0.493	Rotor 1 exit (relative)	yes
0.9	0.51857	0.6065	1.166	0.622	Rotor 1 exit (relative)	yes
0.95	0.52866	0.6183	1.229	0.71	Rotor 1 exit (relative)	yes
1	0.53825	0.6296	1.296	0.772	Rotor 1 exit (relative)	yes
1.05	0.54733	0.6402	1.367	0.815	Rotor 1 exit (relative)	yes
1.1	0.5558	0.6501	1.441	0.845	Rotor 1 exit (relative)	yes
1.15	0.56356	0.6592	1.519	0.867	Rotor 1 exit (relative)	yes
1.2	0.57046	0.6672	1.6	0.882	Rotor 1 exit (relative)	yes

Stall onset (Howell)

N/N_des	m_stall [kg/s]	PR	Eff	Stalling row
0.5	0.26958	1.116	0.934	Rotor 1
0.55	0.29412	1.142	0.934	Rotor 1
0.6	0.31798	1.171	0.935	Rotor 1
0.65	0.34111	1.203	0.935	Rotor 1
0.7	0.36346	1.239	0.936	Rotor 1
0.75	0.38497	1.278	0.936	Rotor 1
0.8	0.40559	1.321	0.936	Rotor 1
0.85	0.42529	1.369	0.937	Rotor 1
0.9	0.444	1.421	0.938	Rotor 1
0.95	0.4617	1.478	0.938	Rotor 1
1	0.47834	1.541	0.939	Rotor 1
1.05	0.49388	1.61	0.94	Rotor 1
1.1	0.5083	1.686	0.94	Rotor 1
1.15	0.52154	1.769	0.941	Rotor 1
1.2	0.5336	1.86	0.942	Rotor 1

Peak (zero slope)

N/N_des	m_peak [kg/s]	PR	Eff	m_stall - m_peak [kg/s]	First limit on throttling
0.5	0.26544	1.116	0.922	0.00414	stall onset (Howell)
0.55	0.28975	1.142	0.922	0.00438	stall onset (Howell)
0.6	0.31342	1.171	0.923	0.00456	stall onset (Howell)
0.65	0.33641	1.203	0.923	0.00471	stall onset (Howell)
0.7	0.35866	1.239	0.924	0.0048	stall onset (Howell)
0.75	0.38013	1.278	0.925	0.00484	stall onset (Howell)
0.8	0.40077	1.322	0.925	0.00483	stall onset (Howell)
0.85	0.42053	1.369	0.926	0.00476	stall onset (Howell)
0.9	0.43936	1.422	0.927	0.00465	stall onset (Howell)
0.95	0.45722	1.479	0.928	0.00448	stall onset (Howell)
1	0.47407	1.542	0.929	0.00427	stall onset (Howell)
1.05	0.48987	1.611	0.931	0.00401	stall onset (Howell)
1.1	0.50458	1.687	0.932	0.00372	stall onset (Howell)
1.15	0.51816	1.77	0.933	0.00339	stall onset (Howell)
1.2	0.53057	1.861	0.935	0.00303	stall onset (Howell)

Koch limit

N/N_des	m_Koch [kg/s]	PR	Eff	Limiting stage
0.5	0.28136	1.113	0.954	Rotors R1+R2
0.55	0.30846	1.137	0.955	Rotors R1+R2
0.6	0.33504	1.163	0.956	Rotors R1+R2
0.65	0.36097	1.192	0.956	Rotors R1+R2
0.7	0.38618	1.224	0.955	Rotors R1+R2
0.75	0.41052	1.257	0.954	Rotors R1+R2
0.8	0.43384	1.293	0.953	Rotors R1+R2
0.85	0.456	1.332	0.95	Rotors R1+R2
0.9	0.47688	1.373	0.947	Rotors R1+R2
0.95	0.49632	1.416	0.943	Rotors R1+R2
1	0.51419	1.461	0.939	Rotors R1+R2
1.05	0.53038	1.508	0.934	Rotors R1+R2
1.1	0.54474	1.558	0.928	Rotors R1+R2
1.15	0.55713	1.608	0.921	Rotors R1+R2
1.2	0.56741	1.661	0.913	Rotors R1+R2

Stability limit & Greitzer B

N/N_des	First limit	m_lim [kg/s]	PR	U [m/s]	a_p [m/s]	U/(2a_p)	(V_p/(A_c L_c))_crit	B	Beyond the limit
0.5	stall onset	0.26958	1.116	146.7	345.9	0.212	14.24	NaN	-
0.55	stall onset	0.29412	1.142	161.3	347.1	0.2324	11.85	NaN	-
0.6	stall onset	0.31798	1.171	176	348.5	0.2526	10.03	NaN	-
0.65	stall onset	0.34111	1.203	190.7	349.9	0.2725	8.62	NaN	-
0.7	stall onset	0.36346	1.239	205.4	351.4	0.2922	7.5	NaN	-
0.75	stall onset	0.38497	1.278	220	353.1	0.3115	6.59	NaN	-
0.8	stall onset	0.40559	1.321	234.7	354.9	0.3306	5.85	NaN	-
0.85	stall onset	0.42529	1.369	249.4	356.8	0.3494	5.24	NaN	-
0.9	stall onset	0.444	1.421	264	358.8	0.3679	4.73	NaN	-
0.95	stall onset	0.4617	1.478	278.7	361	0.386	4.29	NaN	-
1	stall onset	0.47834	1.541	293.4	363.2	0.4038	3.92	NaN	-
1.05	stall onset	0.49388	1.61	308	365.6	0.4212	3.61	NaN	-
1.1	stall onset	0.5083	1.686	322.7	368.1	0.4383	3.33	NaN	-
1.15	stall onset	0.52154	1.769	337.4	370.8	0.4549	3.09	NaN	-
1.2	stall onset	0.5336	1.86	352	373.6	0.4712	2.88	NaN	-

Blade rows

Operating point (N = 1)	m [kg/s]	Blade row	Incidence i [deg]	(i - i_des)/deflection	Loss omega	omega/omega_des	M_in (row frame)	M_out (row frame)	de Haller V_out/V_in	Deviation [deg]	Watch
Design point	0.5	Rotor 1	-6.42	0	0.0343	1	0.75	0.604	0.82	12.63	
Design point	0.5	Rotor 2	-4.59	0	0.0286	1	0.581	0.507	0.88	11.16	
First limit (stall onset)	0.47834	Rotor 1	-3.33	0.235	0.0686	2	0.741	0.56	0.773	13.39	Howell stall
First limit (stall onset)	0.47834	Rotor 2	-4.05	0.028	0.0291	1.016	0.537	0.469	0.879	11.22	
Near choke (0.99 m_choke)	0.53287	Rotor 1	-11.79	-0.409	0.1343	3.916	0.772	0.816	1.051	12.62	
Near choke (0.99 m_choke)	0.53287	Rotor 2	-3.72	0.046	0.0299	1.043	0.792	0.64	0.825	11.26	